Benjamin Eichinger

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Research Interests

direct and inverse spectral theory of Jacobi matrices, universality phenomena for orthogonal polynomials and random matrices, Schrödinger operators, canonical systems, integrable systems, extremal problems of Chebyshev type

Education

- **Habilitation in Mathematics**, TU Wien, Austria — 2025
  Thesis: *Eigenvalue asymptotics of finite difference and differential operators*
- **Ph.D. in Mathematics**, Johannes Kepler University Linz, Austria — 2017
  Thesis: *Periodic GMP matrices and asymptotics of extremal polynomials for Chebyshev and Ahlfors problems in the complex plane*
  Advisor: Peter Yuditskii
- **M.Sc. in Mathematics**, Johannes Kepler University Linz, Austria — 2014
- **B.Sc. in Mathematics**, Johannes Kepler University Linz, Austria — 2012

Academic Positions

- **Senior Scientist**, TU Wien, Austria — 2026–present
- **Lecturer**, Lancaster University, UK — 2024–2026
- **Postdoctoral Fellow**, TU Wien, Austria — 2021–2024
  funded by FWF stand-alone project P 33885
- **Postdoctoral Fellow**, Johannes Kepler University Linz, Austria — 2020–2021
  funded by Erwin Schrödinger fellowship J 4138
- **Postdoctoral Fellow**, Rice University, USA — 2019–2020
  funded by Erwin Schrödinger fellowship J 4138
- **Postdoctoral Fellow**, Lund University, Sweden — 2018–2019
  funded by Erwin Schrödinger fellowship J 4138

Awards, and Prizes

- SIAM Gábor Szegő Prize, 2026.
- Research grant from FWF Austrian Science Fund P 33885, “Szegö theorems for semibounded subsets of the real axis”, €326.308, 2021–2024.
- Erwin Schrödinger fellowship awarded from FWF Austrian Science Fund J 4138, “Extremal polynomials on subsets of the unit circle”, €161.390, 2018–2021.

Editorial Service

- Advances in Approximation Theory 2026–

Teaching Experience

Lectures

- **Lebesgue Integration**, Lancaster University, 2026.
- **Further Calculus**, Lancaster University, 2025.
- **Further Calculus**, Lancaster University, 2024.
- **Orthogonal Polynomials**, TU Wien, 2023.
- **Introduction to Hardy Spaces**, TU Wien, 2022.
- **Complex Analysis**, JKU Linz, 2020.
- **Calculus II**, Rice University, 2019.

Teaching Assistant

- Functional Analysis, TU Wien, 2022.
- Functional Analysis, Dynamical Systems and Chaos, Probability Theory and Statistics, Analysis, JKU Linz, 2014-2017.

Conferences organized

- Organizer of workshop: *Spectral Theory by the Lakes*, Lancaster, UK, April 2026
- Organizer of conference: *Operator Theory and Approximation*, Vienna, Austria, July 2024
- Organizer of conference: *Complex Analysis, Spectral Theory and Approximation meet in Linz*, Linz, Austria, 2022
- Co-organizer of Mini-Symposium: *Extremal polynomials and almost periodicity* 15th International Symposium on Orthogonal Polynomials, Special Functions and Applications, Hagenberg, Austria, 2019

Invited Talks

- **Prize Lecture**: 18th International Symposium on Orthogonal Polynomials, Special Functions and Applications, Tokyo, Japan, 2026
- **Plenary talk**: Spectral Theory and Mathematical Physics, in honor of Barry Simon's 80th birthday, Pasadena, USA, 2026.
- **Invited talk**: Inverse Spectral Theory Workshop, Emory, 2026.
- **Invited section talk**: 77th British Mathematical Colloquium, Cardiff, UK, 2026.
- **Plenary talk**: Extremal Polynomials and Dynamical Systems, Copenhagen, Denmark, 2025.
- **Invited talk**: North British Functional Analysis Seminar, Glasgow, UK, 2025.
- **Invited section talk**: International Congress on Mathematical Physics, Strasbourg, France, 2024

- **Plenary talk**: 17th International Symposium on Orthogonal Polynomials, Special Functions and Applications, Granada, Spain, 2024
- **Plenary talk**: International Conference on Spectral Theory and Approximation, Lund, Sweden, 2023.
- **Plenary talk**: Contemporary Analysis and Its Applications, Portoroz, Slovenia, 2023.
- **Invited talk**: Øresund Seminar, Lund, Sweden, 2022.
- **Invited talk**: Workshop on Reflectionless Operators, Oberwolfach, Germany, 2017.
- **Invited talk**: 35th Annual Western States Mathematical Physics Meeting, Pasadena, California, 2017

Preprints

26. *Chebyshev polynomials on a Jordan arc*, with B. Buchecker and O. Rubin and A. Wennman, arXiv:2608.13445, 44 pages.

25. *New universality classes associated to fractals*, with A. Kheifets and M. Lukić and P. Yuditskii, arXiv:2607.28379, 63 pages.

24. *Clock spacing for two-sided Jacobi matrices*, with M. Lukić and G. Young, arXiv:2606.10134, 19 pages.

23. *Weighted residual polynomials on a circular arc*, with B. Christiansen and O. Rubin and M. Zinchenko, preprint, arXiv:2602.05428, 32 pages.

22. *Necessary and sufficient conditions for universality limits*, with M. Lukić and H. Woracek, preprint, arXiv:2409.18045, 78 pages.

21. *Homogeneous spaces of entire functions*, with H. Woracek, preprint, arXiv: 2407.04979, 51 pages.

20. *Stahl–Totik Regularity for Dirac Operators*, with E. Gwaltney and M. Lukić, preprint, arXiv:2012.12889, 34 pages.

19. *Asymptotics of L^r extremal polynomials for $0 < r \leq \infty$ on C^{1+} Jordan regions*, with B. Buchecker and M. Zinchenko, preprint, arXiv:2502.17616, 26 pages.

Publications

18. *An approach to universality using Weyl m-functions*, with M. Lukić and B. Simanek, **Ann. of Math. (2)**, 203, (2026) 40 pages.

17. *On point spectrum of Jacobi matrices generated by iterations of quadratic polynomials*, with M. Lukić and P. Yuditskii, **Comm. Math. Phys.**, 407, (2026), 20 pages.

16. *A Weyl matrix perspective on unbounded non self-adjoint Jacobi matrices*, with M. Lukić and G. Young, **Complex Anal. Oper. Theory**, 19, (2025), 30 pages.

15. *Extremal polynomials and polynomials preimages*, with J.S. Christiansen and O. Rubin, **Constr. Approx.**, 63 (2025), 523–563.

14. *Stahl–Totik regularity for continuum Schrödinger operators*, with M. Lukić, **Anal. PDE**, 18, (2025), 591–628.

13. *Asymptotics of Chebyshev rational functions with respect to subsets of the real line*, with M. Lukić and G. Young, **Constr. Approx.**, 59 (2024), no. 3, 541–581.

12. *Asymptotics for Christoffel functions associated to continuum Schrödinger operators*, **J. Anal. Math.**, 153 (2024), no. 2, 519–553.

11. *Orthogonal rational functions with real poles, root asymptotics, and GMP matrices*, with M. Lukić and G. Young, **Trans. Amer. Math. Soc.**, 10, (2023), 1–47.

10. *Limit-Periodic Dirac Operators with Thin Spectra*, with J. Fillman and E. Gwaltney and M. Lukić, **J. Funct. Anal.**, 283 (2022), no. 12, 1–35.

9. *Pointwise Remez inequality*, with P. Yuditskii, **Constr. Approx.**, 54, (2021), 529–554

8. *Spectral properties of Schrödinger operators associated to almost minimal substitution systems* , with P. Gohlke, **Ann. Henri Poincaré**, 22, (2021), 1377–1427.

7. *Szegő's Theorem for Canonical Systems: the Arov Gauge and a Sum Rule*, with D. Damanik and P. Yuditskii, **J. Spectr. Theory**, 11, (2021),1255–1277.

6. *Finite-gap CMV matrices: Periodic coordinates and a Magic Formula*, with J.S. Christiansen and T. VandenBoom, **Int. Math. Res. Not.**, (2020), 1–70

5. *KdV hierarchy via Abelian coverings and operator identities*, with T. VandenBoom and P. Yuditskii, **Trans. Amer. Math. Soc. Ser. B**, **6** (2019), 1–44.

4. *Ahlfors problem for polynomials*, with P. Yuditskii, special issues dedicated to the 150th anniversary of **Mat. Sb.**, 209, (2018), no. 3, 34–66.

3. *Szegő-Widom asymptotics of Chebyshev polynomials on Circular Arcs*, **J. Approx. Theory**, **217**, (2017), 15–25.

2. *Periodic GMP matrices*, **SIGMA Symmetry Integrability Geom. Methods Appl.**, 12 (2016), 1–19.

1. *Jacobi Flow on SMP Matrices and Killip-Simon Problem on Two Disjoint Intervals*, with F. Puchhammer and P. Yuditskii, **Compt. Methods Funct. Theory**, 16, (2014), 3–41.